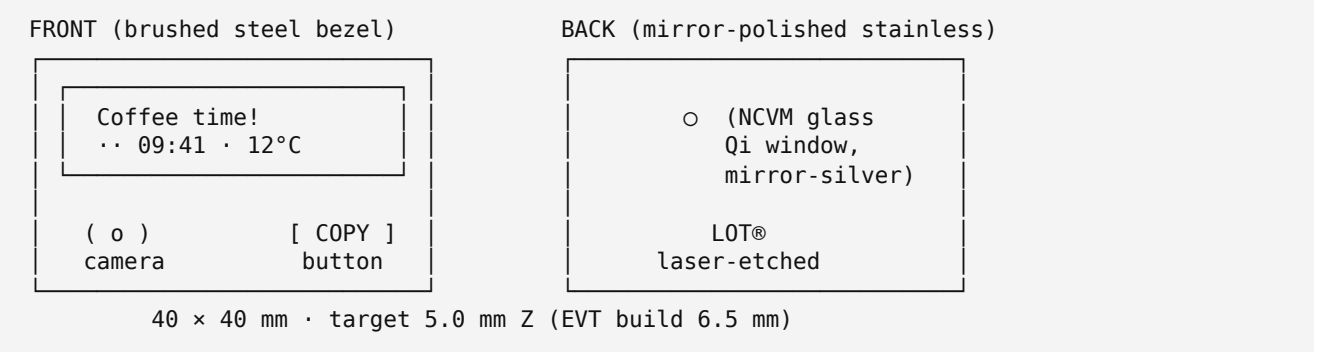
LOT TILE — hardware computer for LOT-systems.com

Codename: LOT TILE · **Part number:** LOT-HW-01 · **Owner:** Vadik, Inventor — COSMO® CIA **Status:** Session 1, plan and EVT design package (2026-11-14). Nothing has been fabricated yet.

A flat, silver, 40 × 40 mm square that acts as a pager for LOT. LOT-systems.com sends short autonomous notifications ("Coffee time!"). The device vibrates once and shows the line on a low-power reflective screen. Pressing the single button (**Copy**) writes that line back to the user's **Log** tab on lot-systems.com. A camera on the same face can attach a photo to the entry. It also carries an AI-grade environmental sensor that reports local weather to the site.



DOCUMENT SET (ONE SUBJECT PER DOCUMENT)

#	Document	Purpose
01	Plan	Product definition, the 19 requirements traced to design decisions
02	Components buying list · bom.csv	Every part, with quantity for 100 units, estimated price, and a buy link
03	Mechanical	Two-part stainless body, Z-stack budget, finishes, sealing
04	Electrical	Block diagram, power tree, pin map, power budget
05	Firmware	Architecture, state machine, OTA, security
06	LOT API connector	Device ↔ lot-systems.com protocol, pairing, Copy → Log
07	Manufacturing: 100 units with PCBWay	PCB, PCBA, CNC steel, assembly, test, certification
08	Roadmap and analysis	Phases, gates, risks, open decisions, budget
09	User manual	End-user guide (also shipped as PDF)

Code:

- `firmware/` — ESP32-S3 PlatformIO project (Arduino framework), EVT skeleton
- `software/lotctl.py` — host software: USB provisioning, pairing, and a device simulator that exercises the LOT API
- `tools/build_pdfs.py` — renders `docs/*.md` into the PDF manuals in `manuals/`

Session reports: `docs/hardware/`.

01 · LOT TILE: Product Plan

Doc: LOT-HW-01-PLAN · **Rev:** A (2026-11-14) · **Author:** Vadik, Inventor, COSMO® CIA (with Claude Code)

1. WHAT IT IS

LOT TILE is a pocket or desk object. It is the physical end of LOT-systems.com.

- The site's AI (Memory Engine, Quantum Intent Engine, contextual prompts) decides when the user should get a nudge and writes it as one short line, for example *"Coffee time!"*, *"Walk. 12°C, clear."* or *"Evening check-in."*
- The tile vibrates once, like a pager, and shows the line.
- **Copy** button, short press: the line is written to the user's **Log** tab on lot-systems.com as a `device_copy` entry. The screen then confirms with "Copied."
- **Copy** button, long press: the camera takes one frame, and the same Log entry gets a photo.
- Every 15 minutes the BME688 sensor reports local temperature, humidity, pressure and air quality. The site uses this in place of, or alongside, the city weather it fetches today (`/api/weather`).

The design follows the LOT style guide: no decorative colour, no emoji, "Done." instead of a check mark, and one primary action.

2. REQUIREMENTS TRACEABILITY (THE 19 POINTS OF THE BRIEF)

#	Brief	Design decision	Doc
1	PCBWay	PCBWay for PCB fabrication, PCBA (turnkey parts), CNC stainless enclosure and mirror polish, from one vendor	07
2	Pager-like notification from the AI site	Server pushes a ≤ 48-character line. Device: LRA haptic pulse, then the line on the Memory LCD	04, 06
3	Two-part stainless body	Part A: back tray (floor and walls, 316L). Part B: front bezel plate (316L) holding the cover glass. Joined with 4 × M1.0 Torx screws and a gasket	03
4	Flat silver square 40 × 40 mm, 5 mm high	40.0 × 40.0 mm footprint. **5.0 mm is achievable only with a chip-down MCU and a custom 1.6 mm cell** (DVT). EVT build is 6.5 mm	03 §3
5	Camera	Ultra-thin FPC VGA module (GC032A / GC0308 class, ≤ 2.5 mm Z), DVP to ESP32-S3	04
6	LOT API connector	New `/api/device/*` namespace with device-token auth and pairing by short code	06
7	Result in PDF manuals	`tools/build_pdfs.py` renders every doc into `manuals/*.pdf`	manuals/
8	Compress the information in each session	(a) Wire format: compact JSON lines ≤ 512 B, with deltas only. (b) Server condenses AI context into one pager line. (c) Each work session ends with a compressed ledger block in the session report	06 §7, report
9	Firmware documents	Firmware architecture, state machine and OTA doc, plus source in `firmware/`	05
10	Software to connect with firmware	`software/lotctl.py`: USB-serial provisioning (Wi-Fi and pairing), log tail, and a device simulator	05 §8
11	Separate documents	One subject per file, 01–09, each also built as its own PDF	README
12	Charger	Qi 5 W charging pad (off-the-shelf in EVT, LOT-branded stainless puck in PVT). USB-C on the pad only	04 §4, 07
13	100-unit run	BOM costed at 100 units. PCBWay PCBA plus CNC lot. FCC path for a pre-certified module	02, 07
14	Weather sensor	Bosch BME688 (T, RH, P, VOC gas) behind a sintered vent with an ePTFE membrane	04
15	AI-grade off-the-shelf sensors	BME688 (on-sensor gas AI via BME AI-Studio), BMI270 IMU (on-chip gesture/wrist-wake AI), LTR-303 ambient light	02, 04

#	Brief	Design decision	Doc
16	Copy button signals the Log tab	`POST /api/device/copy` creates Log `event: 'device_copy'`, which must be added to the Log tab's `displayableEvents`	06 §5
17	One side polished stainless steel	Back: mirror-polished 316L (#8 finish), with the Qi window in NCVM mirror-silver glass so the side still reads as uniform silver	03 §4
18	Other side: camera, screen, button	Front: 1.08" Memory LCD (top), camera (bottom left), Copy key (bottom right)	03 §2
19	Wireless charger	Qi RX, TI BQ51050B (receiver and Li-ion charger in one chip), with a 20 × 20 mm receiver coil behind the back window	04 §4

3. THE ONE PHYSICS CONFLICT TO RESOLVE

Solid stainless steel blocks wireless charging. A Qi field induces eddy currents in a steel back, which heats the steel and stops charging. Points 17 and 19 conflict if the back is one solid steel plate.

Proposed resolution (default, **decision D-1**): cut a Ø 26 mm window in the back plate and fill it with a flush cover of chemically strengthened glass with **NCVM coating** (non-conductive vacuum metallization). This is the finish phones use to look metallic while staying RF and induction transparent. The back still reads as one mirror-silver face, and the window shows only as a thin seam ring.

Alternatives, if Vadik prefers an all-steel back:

Option	Look	Charging	Cost / risk
A. NCVM glass window (default)	Silver, faint seam ring	Qi 5 W (practical 1–2 W)	Low
B. All-steel back, magnetic pogo pins	Pure polished steel, 2 pads visible	Wired via dock, fast	Low, but it is not "wireless"
C. All-steel back, charge face-down through the front glass	Pure back	Coil must sit under the display, which needs custom flex coil and display stack	High
D. Titanium or ceramic back	Different material	Qi works through zirconia	Changes brand material

4. PRODUCT DECISIONS TAKEN IN THIS PLAN

1. **MCU: ESP32-S3.** Wi-Fi and BLE, camera DVP interface, PSRAM for frames, mature Arduino/IDF, low cost. EVT uses the pre-certified `ESP32-S3-MINI-1U/-1-N4R2` module (FCC modular grant, 2.4 mm Z). The 5.0 mm DVT moves to chip-down `ESP32-S3FH4R2` (QFN56 7 × 7 × 0.9 mm). That needs our own RF layout and a full FCC Part 15C grant.
2. **Screen: Sharp Memory LCD, 1.08", 160 × 68 (LS011B7DH03).** Reflective silver-grey look, readable in sun, ~µW static draw (holds the last notification with power off the MCU) and about 1 mm thick. It fits the LOT monochrome aesthetic.
3. **Pager haptic:** 8 mm Z-axis LRA and TI DRV2605L (licensed haptic library, one-line "pager" waveform).
4. **Connectivity:** standalone Wi-Fi with BLE provisioning. No phone app is required in v1. Pairing uses a 6-character code entered on lot-systems.com.
5. **Delivery model:** the device wakes on a timer (default every 5 min, adaptive) and fetches pending notifications. Opening a socket for true push is a v2 option (power cost below).

5. OUT OF SCOPE FOR V1

Microphone and speaker, GPS, cellular, colour display, water resistance above IP54, any on-device generative AI (all AI runs on lot-systems.com).

6. SUCCESS CRITERIA FOR THE 100-UNIT RUN

- ≥ 95 % first-pass yield at final functional test
- ≥ 3 days battery at the default 5-minute cadence (EVT cell) and ≥ 2 days (DVT 5 mm cell)
- Notification latency ≤ cadence + 10 s. Copy-to-Log round trip ≤ 3 s on good Wi-Fi
- Back face passes cosmetic spec: #8 mirror, no visible tool marks at 30 cm under D65 light

02 · Components Buying List (BOM)

Doc: LOT-HW-01-BOM · **Rev:** A (2026-11-14) · Machine-readable copy: `bom.csv`

> **Prices are estimates** at a 100-unit run (with about 10 % overage for SMT attrition and test spares) and were not live-quoted in this session. Before ordering, confirm every line with a PCBWay turnkey quote or a DigiKey/LCSC cart. Links go to the vendor search for the exact manufacturer part number, so they do not break when a product page moves.

EVT / PVT BUILD (6.5 MM, PRE-CERTIFIED MODULE)

Ref	Part	MPN	Qty/unit	Order (100 run)	Est. $/unit	Est. line $	Buy
U1	ESP32-S3 module 4MB flash 2MB PSRAM (EVT)	`ESP32-S3-MINI-1-N4R2`	1	110	$3.20	$352.00	LCSC/DigiKey
DS1	Sharp Memory LCD 1.08in 160x68	`LS011B7DH03`	1	105	$9.50	$997.50	Mouser/DigiKey
J1	0.5mm FPC connector 10-pin back-flip	`Hirose FH34SRJ-10S-0.5SH`	1	110	$0.60	$66.00	DigiKey
CAM1	Ultra-thin VGA camera module DVP FPC (<=2.5mm Z)	`GC032A / GC0308 FPC module`	1	110	$3.00	$330.00	LCSC/Alibaba module houses
J2	0.5mm FPC connector 24-pin	`Hirose FH34SRJ-24S-0.5SH`	1	110	$0.80	$88.00	DigiKey
U2	Environmental sensor T/RH/P/gas with AI-Studio	`Bosch BME688`	1	110	$7.80	$858.00	DigiKey/Mouser
U3	6-axis IMU with on-chip gesture/wrist-wake AI	`Bosch BMI270`	1	110	$2.60	$286.00	DigiKey
U4	Ambient light sensor	`Lite-On LTR-303ALS-01`	1	110	$0.70	$77.00	DigiKey
U5	Qi receiver + Li-ion charger single chip	`TI BQ51050BRHLR`	1	110	$3.10	$341.00	DigiKey/TI
L1	Qi receiver coil ~20x20mm thin with ferrite	`TDK WR202010-18M8-G (or Wurth WE-WPCC equiv)`	1	110	$2.50	$275.00	DigiKey

Ref	Part	MPN	Qty/unit	Order (100 run)	Est. $/unit	Est. line $	Buy
U6	Ultra-low-Iq buck 3.3V 750mA	TI TPS62840DLCR	1	110	$0.95	$104.50	DigiKey
U7	Fuel gauge ModelGauge	Analog MAX17048G+T10	1	110	$1.40	$154.00	DigiKey
U8	Battery protection IC	ABLIC S-82D1AAA-A8T2U7 (or DW01 equiv)	1	110	$0.35	$38.50	LCSC
BT1	LiPo cell EVT 3.0x30x30mm ~150mAh with PCM	303030 class	1	110	$2.20	$242.00	Alibaba (Grepow/PKCELL/EEMB)
U9	Haptic driver (LRA/ERM licensed library)	TI DRV2605LDGSR	1	110	$1.60	$176.00	DigiKey
M1	Z-axis LRA 8mm dia x 2.5mm	Jinlong G0825001 / Vybronics VG0825001	1	110	$1.80	$198.00	DigiKey
SW1	Low-profile tact switch (Copy)	C&K KMR211NG LFS (or Alps SKRPABE010)	1	110	$0.45	$49.50	DigiKey
J3	Tag-Connect TC2030 footprint (no part)	TC2030-IDC-NL footprint	1	0	$0.00	$0.00	Tag-Connect
PASS	Caps/resistors/ferrites 0201-0402 kit	various	1	110	$0.80	$88.00	PCBWay turnkey
PCB	Main PCB 4-layer 0.6mm 36x36mm HDI ENIG	custom	1	110	$4.00	$440.00	PCBWay
FPC	Flex for camera/antenna/LRA (optional)	custom	1	110	$1.50	$165.00	PCBWay
ENC-A	Back tray 316L CNC + #8 mirror polish + laser LOT mark	custom	1	110	$18.00	$1,980.00	PCBWay CNC
ENC-B	Front bezel 316L CNC + bead-blast/brush	custom	1	110	$12.00	$1,320.00	PCBWay CNC
GL1	Front cover glass 0.4mm chemically strengthened + black mask print	custom	1	110	$2.50	$275.00	Alibaba cover-glass vendors
GL2	Back Qi window 26mm dia 0.5mm glass + NCVM mirror-silver	custom	1	110	$3.00	$330.00	NCVM cover-glass vendors
VENT	ePTFE adhesive vent membrane 3mm	Gore PolyVent / Donaldson Tetratex equiv	1	110	$0.60	$66.00	DigiKey
GSK	Die-cut gasket + PSA (3M 300LSE / VHB)	custom die-cut	1	110	$0.50	$55.00	PCBWay / local die-cut
SCR	M1.0 x 2.5 Torx T3 stainless screws	custom/standard	4	450	$0.06	$27.00	McMaster/Alibaba

Ref	Part	MPN	Qty/unit	Order (100 run)	Est. $/unit	Est. line $	Buy
KEY	Steel Copy key cap + silicone plunger	`custom`	1	110	$1.20	$132.00	PCBWay CNC + silicone vendor
CHG	Qi 5W charging pad (EVT off-the-shelf)	`any WPC-certified 5W pad`	1	105	$4.00	$420.00	Alibaba/Amazon
CBL	USB-C cable 1m braided	`generic`	1	105	$1.20	$126.00	Alibaba
PKG	Box + insert + printed quick-start card	`custom`	1	105	$3.50	$367.50	PCBWay / print vendor

Estimated material cost per unit: $95.59 · **Estimated material for the 100-unit run: $10,424.50**

Not included above: PCBA labour and stencil (07 §3), CNC set-up/NRE, fixtures, certification and freight. See the full run budget in 08 · Roadmap §5.

DVT SUBSTITUTIONS FOR THE 5.0 MM TARGET

Ref	Part	MPN	Est. $/unit	Buy	Notes
U1-DVT	ESP32-S3 SoC QFN56 4MB flash 2MB PSRAM (DVT 5mm)	`ESP32-S3FH4R2`	$2.40	LCSC	needs 40MHz XTAL + chip antenna + own FCC grant
Y1	40MHz crystal 2016 (DVT only)	`TXC 8Y40000013`	$0.15	LCSC	DVT only
BT1-DVT	Custom ultra-thin LiPo 1.6x30x30mm ~70mAh	`custom (Grepow / EEMB)`	$4.50	Grepow	custom MOQ often 500-1000; quote at 100 with NRE
ANT1	2.4GHz chip antenna (DVT only)	`Johanson 2450AT18B100`	$0.40	DigiKey	EVT module has PCB antenna -> needs non-metal window too

ORDERING SEQUENCE

1. **Long lead and custom first (week 0):** cover glass (GL1), NCVM window (GL2), LiPo cell (BT1, needs UN38.3 docs), camera module (CAM1, request a Z-height drawing from 3 vendors).
2. **Dev kits now (week 0, about $150):** ESP32-S3-DevKitC-1-N8R8, Adafruit Sharp Memory display breakout (LS011B7DH03-based, product 3502 class), Pimoroni/Adafruit BME688 breakout, Adafruit DRV2605L breakout with LRA, TI BQ51050BEVM-764 Qi receiver EVM, any Qi 5 W pad.
3. **PCBA turnkey (week 4):** upload Gerbers, the centroid file and this BOM to PCBWay. Let PCBWay source the passives and commodity ICs. Consign the display, camera and cell.
4. **CNC (week 4, in parallel):** 5 sample enclosures first (finish check), then 110.

DEV-KIT BREADBOARD LIST (EVT-0, NO CUSTOM PCB)

Item	Why	Link
ESP32-S3-DevKitC-1-N8R8	MCU, Wi-Fi, PSRAM, camera pins	https://www.digikey.com/en/products/result?keywords=ESP32-S3-DEVKITC-1-N8R8
Sharp Memory LCD breakout	Screen and driver code	https://www.adafruit.com/search?q=sharp+memory+display
BME688 breakout	Weather and AI gas	https://www.digikey.com/en/products/result?keywords=BME688+breakout

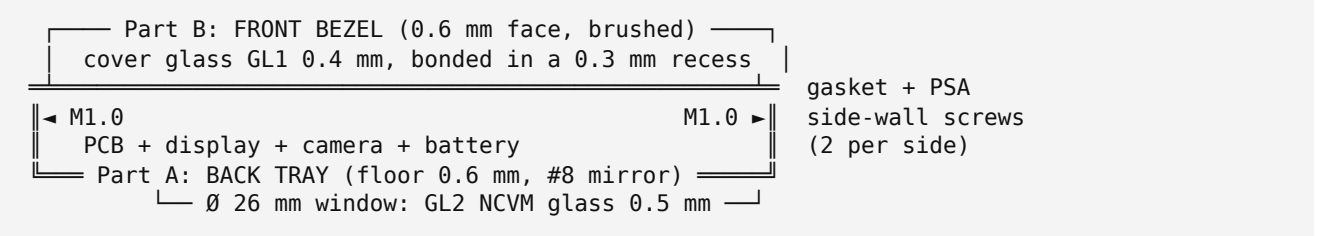
Item	Why	Link
DRV2605L haptic breakout and LRA	Pager buzz	https://www.adafruit.com/search?q=DRV2605L
BMI270 breakout	Pickup-to-wake	https://www.digikey.com/en/products/result?keywords=BMI270+shuttle
OV2640/GC032A camera on FPC	Capture path	https://www.lcsc.com/search?q=GC032A
TI BQ51050BEVM-764	Qi receive and charge	https://www.ti.com/tool/BQ51050BEVM-764
150 mAh 303030 LiPo	Battery	https://www.adafruit.com/search?q=lipo+150mAh

03 · Mechanical: Two-Part Stainless Body

Doc: LOT-HW-01-MECH · **Rev:** A (2026-11-14)

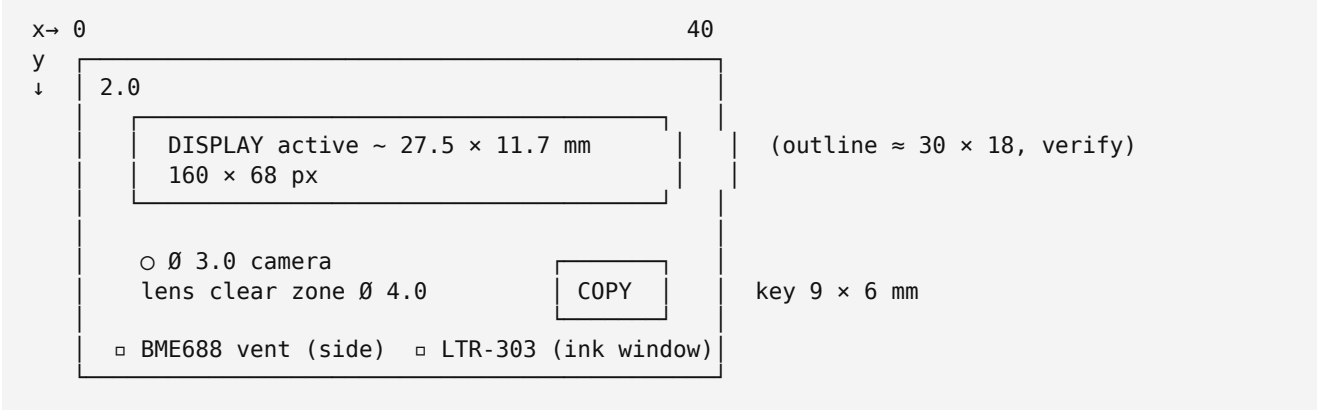
1. ARCHITECTURE

Two machined 316L stainless parts (304 is acceptable for EVT, quote both):



- **Part A, back tray:** floor and four walls machined from one block. Outer corner radius R4, edge chamfer 0.3 mm. Mirror polish #8 on the floor. Walls are brushed in the vertical direction. Laser-etched LOT® mark and serial on the floor, which is *outside*.
- **Part B, front bezel plate:** a 40 × 40 mm plate with apertures for the display window, camera and Copy key. The cover glass covers the display and camera. The Copy key is a separate steel cap in its own pocket.
- **Joining:** 4 × M1.0 × 2.5 Torx T3 screws run *horizontally* through two opposite side walls (2 per side), countersunk and flush, into tapped tabs on the bezel. No screw heads appear on either face, so the mirror back and the front stay clean.
- **Seal:** die-cut 0.15 mm gasket with PSA between the parts. Target IP54 (splash and dust).

2. FRONT FACE LAYOUT (40 × 40 MM)



The camera and the ALS sit under the cover glass, which has an IR-transparent ink window for the ALS and a clear hole in the print for the camera. The BME688 vents through a 0.8 mm hole in the side wall, sealed with an ePTFE membrane, so the reading reflects ambient air, not air trapped in the case.

3. Z-STACK BUDGET: 5.0 MM IS A STRETCH TARGET

Layer	EVT (module) mm	DVT (chip-down) mm	Notes
Back floor (316L)	0.60	0.50	≤ 0.5 mm risks oil-canning on a 40 mm span. Add a rib
Qi coil + ferrite (TDK WR202010)	0.90	0.70	Thinner Würth or Sunlord coils exist. Verify efficiency
Battery	3.00 (303030)	1.60 (custom)	Largest single lever
Air / swell allowance	0.20	0.10	LiPo swells ~8 % over its life
PCB 4-layer	0.60	0.40	0.4 mm is HDI-class at PCBWay
Tallest component on PCB	2.40 (S3-MINI module)	0.90 (QFN56)	The LRA (2.5 mm) and camera (≤ 2.5 mm) sit in a PCB **cut-out** beside the battery, not on top of it
Display (Memory LCD)	0.95	0.95	Mounted to the bezel. Overlaps the module height zone, only if beside it
Cover glass + front plate	0.60	0.50	
Total (with parallel stacking)	**≈ 6.5**	**≈ 5.0**	The components that run in parallel (camera, LRA) are not summed

Conclusion. 5.0 mm needs a chip-down ESP32-S3, a custom 1.6 mm cell and a PCB with cut-outs. EVT is planned at 6.5 mm so the electronics can be validated with off-the-shelf, pre-certified parts. DVT then thins it. (Decision **D-2**.)

4. FINISHES AND THE SILVER LOOK

Surface	Finish	Spec
Back floor	Mirror #8	Ra ≤ 0.05 µm, no orange-peel, no swirl visible at 30 cm
Qi window GL2	NCVM mirror silver on 0.5 mm strengthened glass	Sheet resistance > 10 MΩ/□ (non-conductive), colour matched to polished 316L, ΔE ≤ 2
Walls	Fine brushed (#4)	Vertical grain
Front bezel	Bead-blasted satin or brushed	Contrast to the mirror back
Laser marks	Black anneal marking (no ablation)	LOT® mark, "LOT TILE", serial and FCC ID on the back floor

5. RF WINDOWS: REQUIRED FOR WI-FI

A steel box is a Faraday cage. The Wi-Fi antenna (the module's PCB antenna in EVT, the chip antenna in DVT) **must sit under the front cover glass**, in the band between the display and the camera, with ≥ 3 mm keep-out from the steel. Plan an antenna tuning pass on EVT units in the closed enclosure. The Qi window on the back is a second, smaller RF opening.

6. THERMAL AND SAFETY

- Qi heats steel near the coil. The window must be ≥ 3 mm larger than the coil's outer diameter, and the BQ51050B NTC must read the coil area. Charge is stopped above 45 °C at the cell.
- The battery sits in a die-cut PET/foam pocket with a 0.2 mm swell gap. It never touches a screw boss.

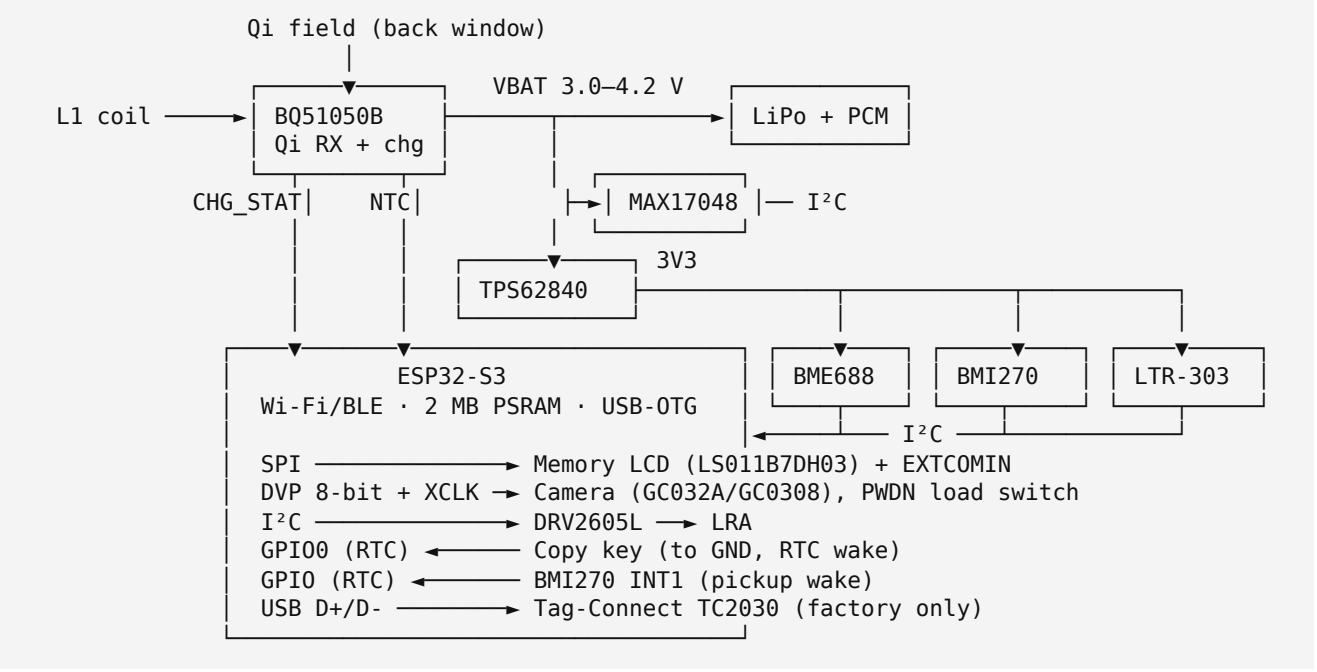
7. CAD DELIVERABLES (NEXT SESSION)

- `cad/lot-tile-back-tray.step`, `cad/lot-tile-front-bezel.step`, `cad/lot-tile-assy.step`
- 2D drawings with GD&T for PCBWay CNC: flatness of the back floor 0.03 mm, wall perpendicularity 0.05 mm
- Cover-glass drawing with print layers (black mask, IR ink window, camera clear zone)

04 · Electrical Design

Doc: LOT-HW-01-ELEC · **Rev:** A (2026-11-14)

1. BLOCK DIAGRAM



2. PIN MAP (ESP32-S3, EVT)

Function	GPIO	Notes
I²C SDA / SCL (sensors, haptic, gauge)	8 / 9	400 kHz, 4.7 kΩ pull-ups on the 3V3 sensor rail
LCD SCLK / MOSI / CS	12 / 11 / 10	Memory LCD: CS is active-high, LSB-first
LCD EXTCOMIN	13	1 Hz toggle from the ULP/LEDC during sleep (VCOM inversion)
LCD DISP	14	Display on/off
CAM D0–D7	15,16,17,18,39,40,41,42	DVP
CAM PCLK / VSYNC / HREF / XCLK	4 / 5 / 6 / 7	XCLK 20 MHz from LEDC
CAM SIOD / SIOC	shares I²C 8 / 9	SCCB-compatible
CAM PWDN (load switch EN)	21	Camera fully unpowered when idle
Copy key	0	RTC GPIO wake, also BOOT strap. Hold at reset = download mode
IMU INT1	1	RTC wake on pickup
Qi CHG_STAT / PGOOD	2 / 3	Charging UI, and ext0 wake on dock
Fuel gauge ALRT	47	Low-battery interrupt
USB D- / D+	19 / 20	Native USB-Serial/JTAG for provisioning (`lotctl`)

3. POWER BUDGET AT THE DEFAULT 5-MINUTE CADENCE

State	Current	Time per hour	mAh per hour
Deep sleep (RTC + ULP EXTCOMIN + sensors idle + LCD static)	18 µA	3 590 s	0.018
Wi-Fi wake, fetch (fast reconnect with a cached BSSID/channel, TLS session resumption)	95 mA avg	12 × 0.9 s	0.285
BME688 forced-mode read, every 15 min	1 mA	4 × 1 s	0.001
Notification: haptic 300 ms + LCD write	80 mA	~3 × 0.4 s	0.027
Copy press: wake + POST	95 mA	~2 × 1.2 s	0.063
Total			**≈ 0.39 mAh per hour ≈ 9.5 mAh per day**

- EVT cell, 150 mAh × 0.85 usable ≈ **13 days** in the ideal case. Plan on **≥ 5 days** with real Wi-Fi retries.
- DVT cell, 70 mAh ≈ **6 days** ideal, ≥ 2–3 days real.
- A persistent socket for true push (v2) costs about 2–3 mA average with DTIM-3 modem sleep, which is about 1 day on the DVT cell. That is why v1 polls.
- Adaptive cadence: 2 minutes while the user is active on lot-systems.com (`next_poll_s` comes from the server), 30 minutes overnight per the user's circadian phase.

4. CHARGING

- **BQ51050B:** WPC 1.1 Qi receiver with an integrated Li-ion charger (4.20 V, set the ILIM resistor to about 150 mA = 1 C for the EVT cell and 70 mA for DVT). The TS/CTRL pin is wired to a 10 kΩ NTC on the cell.

- **Charger accessory:** EVT ships with an off-the-shelf WPC-certified 5 W pad. PVT ships with a LOT-branded stainless puck that has a zirconia top, a magnet ring for alignment (the steel walls of the device are only weakly attracted, so use a thin steel shim plate behind GL2) and USB-C input.
- The screen shows `Charging. 64%` and then `Charged.` (LOT style, with a period).

5. SENSORS: "AI-GRADE, OFF-THE-SHELF"

Sensor	AI feature used	Output to LOT
Bosch BME688	BSEC2 library (IAQ index), plus a custom gas-class model trained in BME AI-Studio (e.g. "coffee brewing", "fresh air", "stale room")	`temp_c, rh, hpa, iaq, gas_class` every 15 min
Bosch BMI270	On-chip wrist-wear and any-motion features, used for pickup-to-wake and double-tap = dismiss	Wake events only. No raw motion leaves the device
Lite-On LTR-303	Lux used for a bedtime "quiet" window	`lux` (coarse) with each telemetry packet

The BME688 gas class gives LOT a context signal: "coffee brewing" detected leads the site to log a coffee moment and suppress the "Coffee time!" nudge.

6. SCHEMATIC AND LAYOUT TASKS (NEXT SESSION)

1. KiCad 8 project `hardware/lot-tile/pcb/`: schematic for all of the above, with ERC clean.
2. Layout: 36 × 36 mm, 4 layers (signal, GND, power, signal), 0.6 mm, ENIG, with cut-outs for the LRA and camera.
3. Antenna keep-out per Espressif's ESP32-S3-MINI-1 hardware design guide, placed on the front-glass edge.
4. Test points for bed-of-nails: VBAT, 3V3, USB, EN, GPIO0, I²C, CHG_STAT.

05 · Firmware

Doc: LOT-HW-01-FW · **Rev:** A (2026-11-14) · **Source:** `../firmware/`

1. STACK

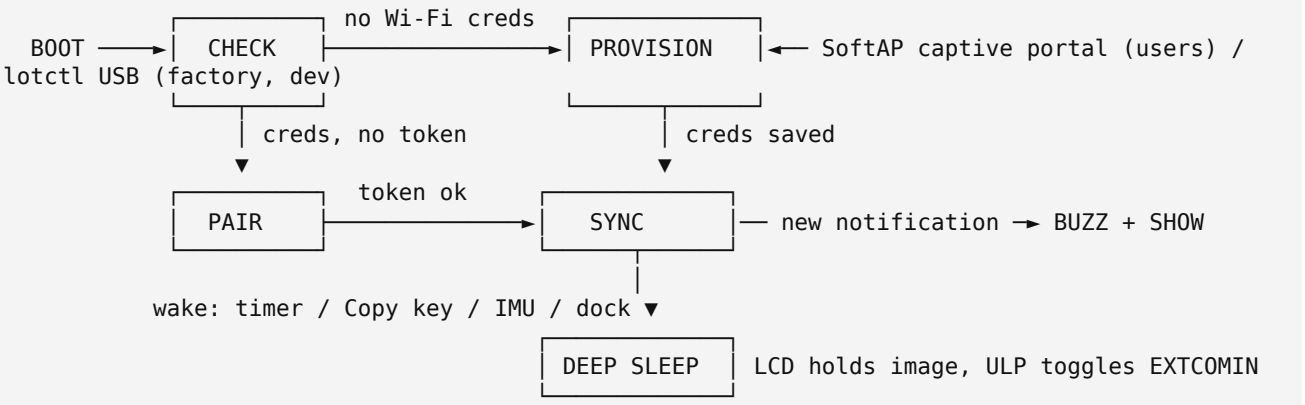
- **Target:** ESP32-S3 (EVT: ESP32-S3-MINI-1-N4R2, dev: ESP32-S3-DevKitC-1-N8R8)
- **Framework:** Arduino-ESP32 3.x on ESP-IDF 5.x, built with PlatformIO (`firmware/platformio.ini`)
- **Libraries:** `esp32-camera`, Adafruit SHARP Memory Display + GFX, Bosch BSEC2 (BME688), SparkFun BMI270, Adafruit DRV2605, ArduinoJson 7, SparkFun MAX1704x

2. MODULE LAYOUT

```
firmware/
├── platformio.ini
├── include/
│   ├── config.h        pins, cadence, API host, limits
│   └── lot_api.h       API client interface
└── src/
    ├── main.cpp        boot → state machine → deep sleep
    ├── lot_api.cpp     HTTPS client: register, pair, sync, copy, photo, telemetry
    ├── ui.cpp          Memory LCD rendering (LOT typography, period style)
    ├── haptic.cpp      DRV2605L "pager" pattern
    ├── sensors.cpp     BME688/BSEC2, LTR-303, fuel gauge
    ├── camera.cpp      single-frame JPEG capture with power gating
    └── provision.cpp   USB-serial line protocol for lotctl
```

Session 1 ships `main.cpp`, `config.h`, `lot_api.h/.cpp` and the provisioning protocol. The other modules are stubbed inside `main.cpp` with TODO markers.

3. STATE MACHINE




```
 Copy short press: POST /copy for the currently shown nid → "Copied."  (no notification:
"Nothing to copy.")
 Copy long press (≥ 800 ms): camera frame → POST /photo → "Copied. Photo."
 Copy hold 10 s: forget token → PAIR
```

Provisioning paths. End users have no USB port (the Tag-Connect pads are internal), so v1 adds a SoftAP captive portal: the AP is named `LOT-TILE-<last 4 of serial>`, the page is served at `192.168.4.1`, and it times out after 10 minutes. `lotctl` over USB-Serial/JTAG is for factory test and development. BLE provisioning through Web Bluetooth on lot-systems.com is a v2 option (it is not supported in iOS Safari). *Session 1 firmware implements the `lotctl` path only. The SoftAP portal is a phase-2 task.*

4. WAKE SOURCES

Source	ESP32-S3 mechanism	Action
Timer	`esp_sleep_enable_timer_wakeup(next_poll_s)`	SYNC. Telemetry every 3rd wake
Copy key (GPIO0)	`ext1` on RTC GPIO, active low	Copy / photo / re-pair
IMU pickup (GPIO1)	`ext1`	Refresh screen with time and weather, quick SYNC
Dock (PGOOD)	`ext1`	Show "Charging." Allow OTA check while on power

5. RENDERING RULES (LOT STYLE)

- One line of notification text, wrapped to at most 2 lines at 160 px. A second, tertiary-weight line shows `HH:MM · 12°C`.
- No icons and no emoji. Status words end with a period: `Copied. Charging. Offline. Paired.`
- The previous notification stays on screen (Memory LCD), so the device reads as "on" with the MCU off.

6. OTA

Two OTA app partitions plus `nvs` and `otadata`. OTA runs only on the dock or at ≥ 50 % battery. Firmware calls `POST /api/device/ota/check`, streams the image, verifies SHA-256 and the ECDSA signature, sets the boot partition, and reboots. If the new image does not reach `SYNC` within 2 boots, the bootloader rolls back (`CONFIG_BOOTLOADER_APP_ROLLBACK_ENABLE`).

7. FACTORY PROGRAMMING AND TEST

1. Bed-of-nails jig with Tag-Connect TC2030 to USB-Serial/JTAG.
2. `esptool.py` flashes the test firmware. The test runs: I²C scan (expect 0x76/0x77 BME688, 0x68/0x69 BMI270, 0x29 LTR-303, 0x5A DRV2605, 0x36 MAX17048), LCD checkerboard, a camera frame-hash check, haptic current draw, Qi charge detect on a pad, and Wi-Fi RSSI against a reference AP.
3. Burn the serial into eFuse user block (`LT01-000001`…`LT01-000100`), then flash production firmware with flash encryption.

8. HOST SOFTWARE: `LOTCTL` (REQUIREMENT 10)

`software/lotctl.py` talks to the firmware over the USB-serial line protocol:

```
> HELLO                        < OK LOT-TILE LT01-000042 fw=0.1.0 hw=EVT1
> WIFI <ssid> <pass>           < OK saved
> API <https://lot-systems.com>   < OK
> STATUS                       < OK wifi=up rssi=-58 paired=yes bat=81 nid=…
> PAIR                         < OK code=K7Q-2MX
> FORGET                       < OK
> LOG                          < (streams firmware log lines until Ctrl-C)
```

`lotctl simulate` runs the same API flow as the firmware, from a laptop, with no hardware. Use it to build and test the server routes before EVT boards exist.

06 · LOT API Connector: Device ↔ lot-systems.com

Doc: LOT-HW-01-API · **Rev:** A (2026-11-14) · **Status:** specification. The server routes are *not yet implemented* (roadmap phase 1).

1. WHY A NEW NAMESPACE

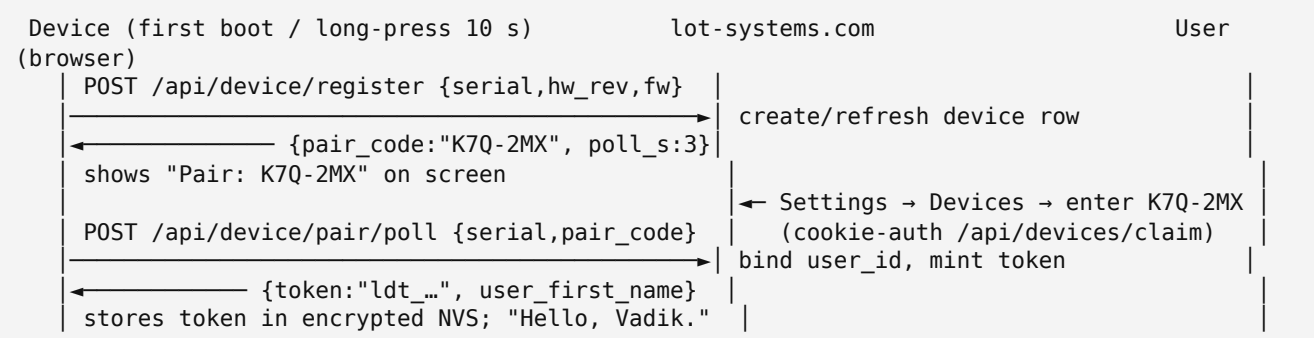
The site authenticates browsers with a JWT in a cookie plus a `Session` row (`src/server/server.ts`, the `onRequest` hook). A device cannot do the email-code login. It needs a long-lived, revocable, per-device token that is scoped to the device endpoints only. The device must never hold a full user session.

New prefix: **`/api/device/*`**, registered beside `/api/public` with its own auth hook.

2. DATA MODEL (NEW MIGRATION)

```
        CREATE TABLE devices (
          id            UUID PRIMARY KEY,
          user_id       UUID REFERENCES users(id) ON DELETE CASCADE,  -- NULL until paired
          serial        TEXT UNIQUE NOT NULL,           -- "LT01-000042", burned at factory
test
          hw_rev        TEXT NOT NULL,                  -- "EVT1"
          fw_version    TEXT,
          token_hash    TEXT,                           -- sha256(device token). The raw
token is never stored
          pair_code     TEXT,                           -- 6 chars, Crockford base32, 10-min
TTL
          pair_expires  TIMESTAMPTZ,
          last_seen_at  TIMESTAMPTZ,
          battery_pct   SMALLINT,
          settings      JSONB DEFAULT '{}',             -- quiet hours, cadence, haptic
strength
          created_at    TIMESTAMPTZ, updated_at TIMESTAMPTZ
        );
        CREATE TABLE device_notifications (
          id           UUID PRIMARY KEY,
          device_id    UUID REFERENCES devices(id) ON DELETE CASCADE,
          user_id      UUID REFERENCES users(id) ON DELETE CASCADE,
          text         VARCHAR(48) NOT NULL,           -- pager line, already compressed
          kind         TEXT NOT NULL,                  -- 'nudge' | 'checkin' | 'weather' |
'system'
          source       JSONB DEFAULT '{}',             -- which engine/prompt produced it
          delivered_at TIMESTAMPTZ, copied_at TIMESTAMPTZ,
          created_at   TIMESTAMPTZ
        );
        CREATE INDEX ON device_notifications (device_id, delivered_at);
```

3. PAIRING (DEVICE-CODE FLOW)



- Token format: `ldt_` + 32 random bytes in base64url. The server stores only `sha256(token)`.
- Revoke from Settings → Devices. Every later device call then gets `401`, and the device returns to pairing.
- Rate limits: register 5/h per IP, poll 1/2 s per serial, claim 10/h per user.

4. ENDPOINTS

All requests use `Authorization: Bearer ldt_…` (except register and pair/poll) and `Content-Type: application/json`. Bodies are ≤ 512 B, except the photo upload.

Method	Path	Body → Response	Purpose
POST	`/api/device/register`	`{serial,hw_rev,fw}` → `{pair_code,expires_s,poll_s}`	Start pairing

Method	Path	Body → Response	Purpose
POST	`/api/device/pair/poll`	`{serial,pair_code}` → `{status:"pending"}` or `{token,name}`	Finish pairing
POST	`/api/devices/claim` (cookie auth)	`{pair_code}` → `{device}`	User binds the device in the browser
GET	`/api/device/sync?since=<cursor>`	→ `{n:[{id,t,k,ts}],cur,next_poll_s,cfg?}`	**The main call.** Pending notifications plus config. Marks them delivered
POST	`/api/device/copy`	`{nid,ts}` → `{log_id}`	**Copy button.** Writes a Log entry
POST	`/api/device/photo`	raw `image/jpeg` body (≤ 60 KB) + header `X-LOT-Notification: <nid>` → `{log_id}`	Long-press Copy. Attaches a photo (no multipart encoder needed on the device)
POST	`/api/device/telemetry`	`{tc,rh,hpa,iaq,gc,lux,bat,rssi}` → `{ok}`	Weather/sensor upload every 15 min
POST	`/api/device/ota/check`	`{fw}` → `{url,sha256,size,sig}` or `{}`	Firmware update

Short keys in `sync`: `t` text, `k` kind, `ts` unix seconds. This keeps responses inside one TLS record.

5. COPY → LOG TAB (REQUIREMENT 16)

`POST /api/device/copy {nid}` does the following on the server:

```
        await fastify.models.Log.create({
          userId: device.userId,
          event: 'device_copy',
          text: notification.text,                          // "Coffee time!"
          metadata: { deviceId: device.id, serial: device.serial, notificationId: nid, kind:
notification.kind },
          context: { source: 'lot-tile', copiedAt: new Date().toISOString(), weather:
lastTelemetry },
        })
```

Two code changes are required so the entry *appears* in the Log tab:

1. Add `'device_copy'` (and `'device_photo'`) to the `LogEvent` union in `src/shared/types/index.ts`.
2. Add both to the `displayableEvents` allow-list in `GET /api/logs` (`src/server/routes/api.ts`, around line 1084). Without this the entry is stored but filtered out of the tab.

Client rendering: show it as a normal Log row with a small `TILE` label in tertiary opacity, matching the LOT style guide.

6. WHERE NOTIFICATIONS COME FROM: THE "AI-POWERED SITE"

A new scheduled job (`src/server/scheduled-jobs.ts`) named **`device-nudge`** runs every 5 minutes for users with a paired device:

1. Gather context the site already computes: local time and circadian phase, Memory Engine answers (for example the morning-beverage answer "Coffee" in `src/server/utils/memory/constants.ts`), recent Log activity, weather, and the latest BME688 telemetry.

2. Decide whether a nudge is due (deterministic rules first, such as "morning beverage window, nothing logged yet"). Call Claude only for wording.
3. **Compress to a pager line**, ≤ 48 characters, in LOT voice: a sentence ending in a period or "!", no emoji. Validate on the server and truncate on a word boundary.
4. Enforce quiet hours, a cap of 8 per day, and a minimum gap of 45 minutes. Insert into `device_notifications`.

Example outputs: `Coffee time!` · `Walk. 12°C, clear.` · `Evening check-in.` · `Air is stale. Open a window.`

7. COMPRESSION (REQUIREMENT 8)

Layer	What is compressed	How
AI → line	Full user context (KBs)	Condensed into one 48-character line on the server
Wire	JSON payloads	Short keys, deltas only (a `since` cursor), ≤ 512 B. No base64 except the photo
Photo	VGA frame	On-device JPEG at q≈12, 320 × 240, ≤ 60 KB
Telemetry	15-min samples	Batched when offline, as a ring buffer of 96 samples (24 h). Sent as one array on reconnect
Session knowledge	Each work session on this project	A "Compressed ledger" block at the end of every session report

8. SECURITY

- TLS 1.2+ only, with the ISRG Root X1 and X2 anchors pinned in firmware (lot-systems.com uses Caddy/Let's Encrypt; see `Caddyfile`).
- The device token lives in NVS encrypted with the flash-encryption key. Secure Boot v2 is enabled at PVT.
- Device endpoints can never read user data beyond its own notifications. They can only write Log rows of type `device_*`.
- OTA images are signed (ECDSA P-256), and the signature is checked before switching boot partitions.

07 · Manufacturing: 100-Unit Run with PCBWay

Doc: LOT-HW-01-MFG · **Rev:** A (2026-11-14)

PCBWay does all four of the main manufacturing steps: PCB fabrication, PCBA (turnkey parts), CNC machining in stainless steel with polishing, and box build. Running all of them through one vendor makes the run simpler to manage.

Service	Link	What we order
PCB fabrication	https://www.pcbway.com/orderonline.aspx	4-layer, 0.6 mm (EVT) / 0.4 mm HDI (DVT), ENIG, 36 × 36 mm, panelised 3 × 3 with mouse bites
Flex PCB (optional)	https://www.pcbway.com/flexible-pcb-manufacturer.html	Camera/antenna/LRA flex, polyimide, 0.1 mm
PCB assembly	https://www.pcbway.com/pcb-assembly.html	Turnkey for commodity parts. Consigned: display, camera, cell

Service	Link	What we order
CNC machining	https://www.pcbway.com/rapid-prototyping/cnc-machining/	316L back tray and front bezel, #8 mirror polish, laser marking
Box build	https://www.pcbway.com/ (quote by email)	Final assembly, test and packaging (PVT)

1. BUILD PHASES AND QUANTITIES

Build	Qty	PCB	Enclosure	Purpose
EVT-0	3	Dev kits on breadboard	3D-printed (SLA)	Firmware and API bring-up
EVT-1	10	Custom PCB, module	5 CNC + 5 SLA	Electrical validation, RF in a steel box, Qi through the window
DVT	20	Chip-down if D-2 = 5 mm	CNC, final finish	Thermal, drop, battery life, cosmetics, pre-scan for FCC
PVT / run	**100 (+10 spares)**	Final	CNC + polish	Customer units

2. FILES PCBWAY NEEDS

- Gerbers (RS-274X) and NC drill, a stack-up note, and impedance control on the antenna feed (50 Ω)
- Centroid / pick-and-place (`.csv`) and assembly drawing (top and bottom)
- BOM in PCBWay format: export `docs/bom.csv` with columns `Designator, Qty, MPN, Manufacturer, Description, Package`
- STEP files for the enclosure plus 2D PDF drawings with tolerances (03 §7)
- Finish spec: "#8 mirror on the outer back face, Ra ≤ 0.05 µm; side walls #4 brushed vertical; no tool marks; protective film before shipping"

3. ESTIMATED RUN COST AT 100 UNITS (USD)

These are estimates for planning, not quotes.

Line	Estimate
Material (02-BOM, EVT variant)	10 425
PCB fab + PCBA labour, stencil, X-ray for QFN/LGA	1 800
CNC NRE / set-up, first-article samples	900
Test fixture (bed of nails, pogo, Tag-Connect, Qi pad)	600
Final assembly + test labour (~15 min/unit)	1 000
Freight (DDP) + battery dangerous-goods surcharge	600
Run subtotal	**≈ 15 300 (≈ $153 per unit)**
FCC (module path: Part 15B + modular grant reuse + SAR exempt check)	3 000 – 5 000
FCC (chip-down path: full Part 15C intentional radiator)	10 000 – 18 000

4. CERTIFICATION AND COMPLIANCE

- **FCC (USA):** the EVT/PVT module path reuses the Espressif modular grant, and the unintentional-radiator test (Part 15B) is still required. The label shows "Contains FCC ID: 2AC7Z-ESPS3MINI1" on the laser-marked back. Re-check the grant for the exact module variant.
- **Qi:** "Qi-compatible" wording only. "Qi Certified" (WPC logo) needs WPC membership and testing, which is not budgeted for 100 units.
- **Battery transport:** UN38.3 test summary and MSDS from the cell vendor, IATA PI 967 (cells inside equipment).
- **RoHS / REACH** declarations from PCBWay and the cell vendor. **Prop 65** check for the California market.
- **WEEE / battery** labels if selling to the EU (out of scope for the first 100).

5. FINAL FUNCTIONAL TEST (EACH UNIT)

1. Program and run the factory test (05 §7). The fixture logs pass/fail to `test-log.csv`, keyed by serial.
2. Qi: place on the reference pad. CHG_STAT must assert within 5 s, and charge current must be within ±15 %.
3. Wi-Fi: associate to the fixture AP. RSSI must be ≥ −65 dBm at 1 m, with the enclosure closed.
4. Camera: capture a target chart and check the JPEG hash range plus mean luminance.
5. Haptic: LRA back-EMF auto-calibration passes on the DRV2605L.
6. Cosmetic: inspect the back mirror under a D65 light booth, then fit the protective film.
7. Ship in "unpaired" state. The first press shows a pair code.

08 · Roadmap and Analysis

Doc: LOT-HW-01-ROAD · **Rev:** A (2026-11-14) · Week 0 = the week Vadik approves decisions D-1 and D-2.

1. PHASES

Phase	Weeks	Deliverables	Exit gate
0 · Plan (done, session 1)	—	Docs 01–09, BOM, API spec, firmware skeleton, `lotctl` with mock server	Vadik signs off D-1…D-5
1 · Server API	0–2	`devices` + `device_notifications` migration, `/api/device/*` routes, Settings → Devices UI, `device_copy` in the Log tab, `device-nudge` job	`lotctl simulate --host https://lot-systems.com` passes against staging
2 · EVT-0 bench	0–3	Dev kits wired. Firmware drivers for LCD, haptic, BME688, camera, sleep. SoftAP Wi-Fi setup portal	"Coffee time!" arrives on the bench unit. Copy appears in Log
3 · EVT-1 design	2–6	KiCad schematic and layout, STEP enclosure, cover-glass drawings	DRC/ERC clean. PCBWay DFM review passes
4 · EVT-1 build	6–9	10 units (PCBWay quick-turn PCBA plus 5 CNC shells)	Power budget measured. RF ≥ −65 dBm closed. Qi charges through GL2
5 · DVT	9–15	Thin stack (if D-2 = 5 mm), 20 units, FCC pre-scan, drop and thermal tests	All of 01 §6 met on ≥ 18 of 20
6 · Certification	13–18	FCC Part 15B (+15C if chip-down), UN38.3 docs	Grant / test report in hand
7 · PVT / 100-unit run	16–22	110 units built, tested, packed. PDF manuals printed and in the box	≥ 95 % first-pass yield

Phase	Weeks	Deliverables	Exit gate
8 · Field	22+	OTA channel, user feedback via Log, v2 planning (push socket, BLE app)	—

Critical path: **custom cell (BT1-DVT) and NCVM window (GL2)**, which have 4–6-week lead times. Order samples in week 0.

2. OPEN DECISIONS FOR VADIK

ID	Decision	Options	Recommendation
D-1	Wireless charging versus an all-steel back	NCVM glass window / all-steel with pogo dock / ceramic back	**NCVM glass window.** It keeps the silver look and Qi works (03 §1, 01 §3)
D-2	Thickness	6.5 mm with module (cheaper, faster, pre-certified) / 5.0 mm chip-down (+$10k FCC, +6 weeks)	Ship the first 100 at **6.5 mm**. Make 5.0 mm the v1.1 run
D-3	Camera purpose	Photo-to-Log only / also QR pairing / disabled until v2	Photo-to-Log on long press. QR pairing is a nice follow-on
D-4	Delivery model	Poll every 5 min (days of battery) / persistent socket (~1 day)	**Poll** with server-driven adaptive cadence
D-5	Steel grade	304 (cheaper, polishes well) / 316L (better corrosion, skin contact)	**316L** for a skin-contact pocket object

3. RISK REGISTER

#	Risk	L	I	Mitigation
R1	Steel enclosure kills Wi-Fi range	H	H	Antenna under the front glass, keep-out, tuning on EVT-1, chip antenna on DVT
R2	Qi heats the steel ring and charging stalls	M	H	Window ≥ coil + 3 mm, NTC at the cell, 1 C limit. Pogo-pin fallback footprint on the PCB
R3	5 mm unachievable with a camera	H	M	D-2: 6.5 mm first. Camera beside the battery in a PCB cut-out
R4	Memory LCD outline does not fit with the camera and button	M	M	Verify the LS011B7DH03 drawing in week 0. Alternative: 0.96" 128 × 64 bare OLED (higher power)
R5	Thin camera module sourcing (single supplier, EOL)	M	M	Qualify 2 module houses. Firmware supports GC032A and GC0308
R6	Custom cell MOQ above 100	H	M	Pay NRE, or use the catalogue 303030 for the first run
R7	FCC delay	M	H	Module path for the first run. Pre-scan at DVT
R8	Notification fatigue	M	M	Server caps: 8/day, 45-min gap, quiet hours, circadian phase
R9	Token leak from a lost device	L	M	Revoke in Settings. Device scope is write-only `device_*` logs plus own notifications

4. ANALYSIS: IS THE CONCEPT SOUND?

- **Strong fit with LOT.** The site already produces proactive, context-aware prompts (Memory Engine questions, self-care, circadian phase). TILE is a *physical, quiet output channel* for them, and the Copy button closes the loop into the Log, which is LOT's record of the day.

- **The hard parts are mechanical and RF, not software.** Steel, a 5 mm thickness and wireless charging pull against each other, and §2 resolves them.
- **The software is low-risk:** about 8 new routes, 2 tables and 1 scheduled job, plus the Log allow-list change. The mock server in `lotctl` already pins the contract.
- **Unit economics at 100:** about $153 build cost plus about 40*perunitamortisedcertification*, *soabout* * *190–$200 landed per unit**. Fine for a limited, numbered "LOT TILE 001/100" edition. At 1 000 units this drops below about $70.

5. BUDGET SUMMARY (100-UNIT EDITION, USD ESTIMATES)

Item	Estimate
Dev kits and EVT-0	300
EVT-1 (10 units, quick-turn)	2 500
DVT (20 units)	4 000
Certification (module path)	4 000
PVT run (07 §3)	15 300
Contingency 15 %	3 915
Total programme	**≈ 30 000**

6. NEXT SESSION (SESSION 2) PLAN

1. Implement phase 1 on the server: migration, `src/server/routes/device-api.ts`, the auth hook, `device_copy` in `LogEvent` and `displayableEvents`, then run `lotctl simulate` against a local server.
2. Start the KiCad schematic in `hardware/lot-tile/pcb/`.
3. Firmware: Memory LCD driver and DRV2605L pager waveform (needs a toolchain-enabled environment to compile).

09 · LOT TILE: User Manual

Doc: LOT-HW-01-UM · **Rev:** A (2026-11-14) · Applies to firmware 0.1.x

IN THE BOX

LOT TILE · charging pad · USB-C cable · this manual

YOUR LOT TILE

- **Front:** screen, camera, **Copy** button.
- **Back:** polished stainless steel. Place this side on the charging pad.

1. CHARGE

Place the tile, back down, on the pad. The screen shows `Charging. 64%`. A full charge takes about 1.5 hours.

2. CONNECT TO WI-FI (FIRST TIME)

1. The screen shows `Wi-Fi: LOT-TILE-4F2A`.
2. On your phone, join the Wi-Fi network **LOT-TILE-4F2A**. A setup page opens. If it does not, go to `http://192.168.4.1`.
3. Choose your home network and enter its password. The tile shows `Connected.` and then a pair code, `Pair: K7Q-2MX`.

3. PAIR WITH LOT-SYSTEMS.COM

1. Open **lot-systems.com → Settings → Devices**.
2. Enter the code from the screen.
3. The tile shows `Hello, <your name>.` and buzzes once. Done.

4. DAILY USE

You see / feel	Meaning
One buzz plus a line, e.g. `Coffee time!`	A notification from LOT
`09:41 · 12°C` under it	The time it arrived and the local temperature
`Offline.`	No Wi-Fi. The tile tries again on its next wake

Copy button

Action	Result
Press	Saves the line on screen to your **Log** on lot-systems.com. The screen shows `Copied.`
Press and hold (about 1 s)	Also takes a photo and attaches it to the same Log entry. The screen shows `Copied. Photo.`
Hold 10 s	Forgets the pairing. The screen shows `Forgotten.` and then a new pair code

Pick the tile up to refresh the screen.

5. QUIET HOURS AND FREQUENCY

Set these in **Settings → Devices** on lot-systems.com: quiet hours, maximum notifications per day, and buzz strength. By default LOT sends at most 8 per day, at least 45 minutes apart, and none during your sleep window.

6. CARE

- The back is mirror-polished 316L stainless steel. Clean it with a microfibre cloth and avoid abrasives.
- The tile is splash-resistant (IP54 target), not waterproof.
- Keep the small vent hole on the side clear. The weather sensor breathes through it.
- Do not charge it above 35 °C ambient. Use only Qi-compatible pads.

7. PRIVACY

- The camera is fully powered off except during a long press.
- Photos go only to your own Log. Sensor readings are temperature, humidity, pressure, air quality class and light level. No audio, location or motion data leaves the device.
- To revoke the tile, go to **Settings → Devices → Remove**. It stops working until it is paired again.

8. SPECIFICATIONS

Size	40 × 40 × 6.5 mm (EVT edition)
Body	Two-part 316L stainless steel, mirror back, brushed walls
Screen	1.08" reflective memory LCD, 160 × 68
Radio	Wi-Fi 2.4 GHz 802.11 b/g/n, Bluetooth LE 5
Sensors	Bosch BME688 (temperature, humidity, pressure, air quality AI), Bosch BMI270 motion, ambient light
Camera	VGA, photo-to-Log
Battery	Li-polymer, about 150 mAh, ≥ 5 days typical
Charging	Qi-compatible wireless, 5 W pad included

LOT® · lot-systems.com · Made by LOT Systems / COSMO® CIA